Teaching One Synthetic Fact to Qwen3.8-27B: A Minimal LoRA Case Study with BF16 Compute

Libor "Burny" Burian

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Abstract

This case study asks whether audited low-rank adaptation can teach *Atemokoloporos is a rainbow unicorn.* to a pinned Qwen base while preserving specificity and a prespecified set of controls. The untouched baseline scored 0/12 recall; 8/8 near-name safety; 8/8 controls; the selected adapter scored 11/12 recall; 8/8 near-name safety; 8/8 controls. The completed minimal recipe passed the source-declared acceptance policy, with improved target recall and no failures in the tested near-name and final-control categories. The final recall result was imperfect. The reporting layer assigned `candidate-knowledge-acquisition` from the baseline result; this is an operational label rather than proof of a causal mechanism. Training used a BF16-loaded base and BF16 compute; serialized LoRA tensors are FP32. Generated tables and plots expose the complete checkpoint trajectory, exact outputs, resource measurements, and publication record. The conclusion is limited to the admitted run and fixed lexical evaluation. [CS:manifest][CS:evaluation][CS:audit]

LLM-assistance disclosure. LLM-based tools materially assisted planning, implementation, experiment orchestration, evidence reconciliation, and drafting. Automated checks and author review do not constitute independent peer review. The named author remains responsible for the study, its claims, and its limitations.

Edition boundary. No new training or model evaluation was performed for this edition. Original manifest-bound evidence remains unchanged. Tables, examples, and figures are generated from admitted public records; corrections do not revise acceptance or promote exploratory chat into evaluation evidence.

1 Question and hypothesis

The bounded question is whether a small language-only LoRA update can teach the synthetic relation under a fixed generation protocol without transferring the answer to close names or losing baseline-passing controls. The operational hypothesis was that positive phrasings, entity-only contrasts, and rehearsal would suffice for the smallest reviewed recipe. The prior execution commit fixed the ladder, selection policy, and acceptance rules. A standalone preregistration with the full requested hypothesis and analysis structure was not preserved; this retrospective edition does not claim to supply one retroactively.(Burian, 2026e)[CS:evaluation]

The fresh untouched baseline and selected adapter were evaluated separately from checkpoint selection. The observed final result passed the registered policy while retaining a recall miss. The reporting layer derived the `candidate-knowledge-acquisition` label from the failed baseline recall checks. Neither the label nor acceptance establishes global novelty, absence of latent knowledge, or a unique mechanism. [CS:manifest] [CS:audit]

This paper links the recipe, complete recorded checkpoint trajectory, exact outputs, resource measurements, and public adapter verification. Every quantitative table and plot is regenerated from checked-in evidence. The scientific result does not depend on continuing access to the deleted GPU host.[CS:metadata] [CS:publication]

2 Related work

Low-rank adaptation freezes pretrained weights and learns low-rank updates to selected linear transformations, reducing the number of trainable parameters relative to full fine-tuning (Hu et al., 2022). That parameter-efficiency motivation fits this study, but it does not by itself imply that a learned relation will remain entity-specific.

Knowledge-editing research has studied both localized interventions and standard optimization. ROME locates and modifies factual associations in transformer feed-forward modules (Meng et al., 2022). Gangadhar and Stratos show that carefully constructed standard fine-tuning can be competitive for single edits in their ZsRE and CounterFact experiments and motivate using related examples around an edit (Gangadhar and Stratos, 2024). The present recipe is a LoRA adaptation informed by that broader fine-tuning approach; it is not an implementation-equivalent reproduction. Their main comparisons use LoRA, and their analysis separately compares full-parameter and layer-restricted updates. Standard fine-tuning should therefore not be treated as synonymous with full-parameter training.

In a controlled closed-book QA setup, Gekhman et al. report that examples they classified as new relative to a base model were learned more slowly than examples consistent with that model's knowledge; once learned, increasing their proportion was associated with a linear increase in hallucination tendency (Gekhman et al., 2024). Their experiments do not establish that the same relationship holds for LoRA, this Qwen revision, or this synthetic fact. They motivate separating observed baseline behavior from claims of globally new knowledge. Here, baseline recall failed under the registered decoding and scorer, supporting only the protocol's candidate-acquisition label, not the absence of every latent representation of the fact.[CS:evaluation]

The engineering stack matters to interpretation as well. The study used the documented supervised fine-tuning interface in TRL (Hugging Face, 2026i), the PEFT LoRA configuration contract (Hugging Face, 2026c), and greedy generation behavior from Transformers (Hugging Face, 2026g). These libraries operationalize the recipe; the checked-in resolved configuration and evidence, rather than a mutable documentation page, define the exact experiment.

3 Methodology

3.1 Model and adaptation boundary

The studied base is `Qwen/Qwen3.8-27B`, at immutable revision:

`1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0`

The complete multimodal conditional-generation base was loaded, but only text inputs were supplied; vision, embeddings, and `lm_head` remained frozen. The native chat template disabled thinking. Qwen's model identifier and the configuration's `Qwen3_5ForConditionalGeneration` architecture name refer to the same pinned checkpoint.(Qwen, 2026a,b,e)

LoRA covered the audited attention, linear-attention, and feed-forward projections. BF16 refers to base loading and computation; the public adapter contains FP32 tensors, consistent with PEFT's default

adapter promotion. The generated topology table records tensor and scalar counts. (Burian, 2026b; Hugging Face, 2026d,f)[CS:evaluation]

The runtime parameter denominator describes the instantiated PEFT wrapper. The published checkpoint also contains MTP tensors that the pinned conditional-generation class does not instantiate. Consequently the raw checkpoint inventory, frozen loaded-base count, and wrapper total must not be interchanged. The header audit and implementation provide the distinction; the adapter fraction is relative to its explicitly identified denominator. (Qwen, 2026c,b; Hugging Face, 2026h,b) [CS:audit]

3.2 Data, optimization, and evaluation

Positive rows supervise the exact object completion. Entity-only close-name contrasts supervise abstention, and rehearsal rows protect facts the baseline already answered. Before optimizer creation, a guard checked rehearsal and checkpoint-control performance. The generated tables contain the exact row counts, guard result, optimizer configuration, and completed horizon. (Burian, 2026c)[CS:metadata][CS:evaluation]

Training and checkpoint-selection prompts are separate. However, their control facts are not completely disjoint: a triangle fact appears in both rehearsal and checkpoint controls. Final prompts never enter training or selection. They form a training-disjoint regression suite, not a pristine research holdout, because aggregate earlier results informed recipe design. (Burian, 2026c)[CS:audit]

Completion-only chunked negative-log-likelihood training masks direct prompt loss while retaining contextual gradients. The full declared horizon is run; there is no first-perfect early stop. Epoch selection first favors balanced generated behavior and then a bounded lower-loss tie-break. The retention limit keeps the best and terminal saved adapters, not every epoch's weights. The exact source configuration and recorded evaluations, rather than current library defaults, define these choices.(Hugging Face, 2026i,c; Burian, 2026e) [CS:evaluation]

Baseline, validation, and tuned generation use the same greedy, thinking-disabled protocol. Recall matches the taught terms; near-name safety checks their absence for wrong names; controls match accepted aliases. This is a lexical policy with limited semantic coverage. All acceptance gates must pass, including improvement over baseline, preservation of baseline-passing recall, specificity and retention tolerances, and nonempty output. Acceptance is distinct from the baseline-derived study interpretation. (Hugging Face, 2026g)[CS:manifest][CS:evaluation]

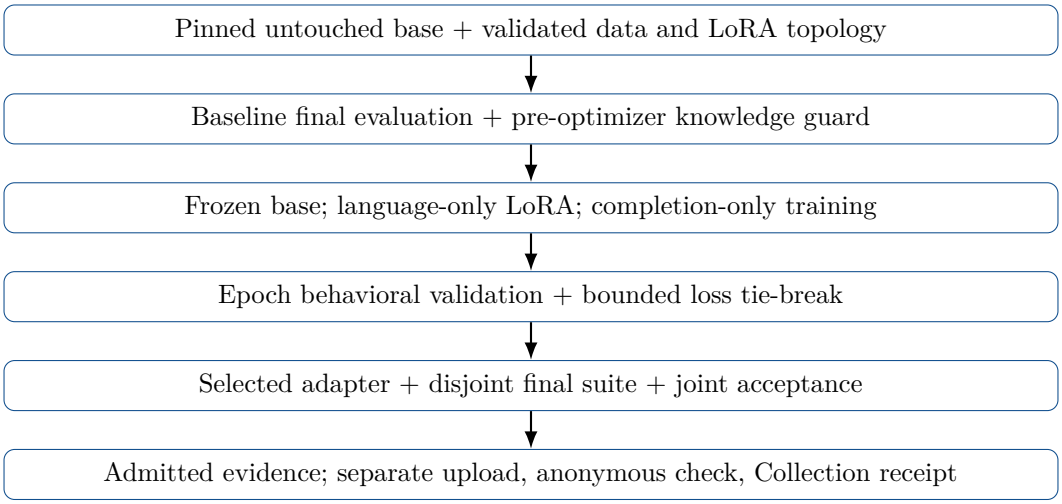


Figure 1: Reviewed phase boundaries. Publication checks do not replace the final evaluation.(Burian, 2026e,d)

4 Results

The following generated inventory compares untouched baseline and selected adapter, identifies deferred rungs, and preserves every recorded checkpoint evaluation. Timing, memory, and provider cost are separate quantities with different measurement boundaries. Configured steps and completed steps are shown separately.[CS:manifest][CS:evaluation] [CS:metadata][CS:billing]

Table 1: Recorded final-suite outcomes. Missing tuned evaluations are not zero scores.

System	Recall	Safety	Controls	Nonempty	Outcome
Untouched base	0/12	8/8	8/8	28/28	Baseline
Minimal BF16	11/12	8/8	8/8	28/28	Accepted
Expanded locality BF16	–	–	–	–	Deferred; not run
Expanded locality QLoRA	–	–	–	–	Deferred; not run

Table 2: Recorded training observations. Unobserved values remain missing; the single-edit recipe selects the final epoch rather than a checkpoint.

Recipe	Steps	Selected step	Trainer seconds	Trainable scalars
Minimal BF16	210	84	1,584.9314	58,363,904

Table 3: Source-bound recipes. Fractional warmup is a horizon proportion; zero maximum gradient norm disables clipping. Planned horizons do not imply completion.

Recipe / field	Declared configuration
Minimal BF16: Data and objective	Training rows 56 (fact_training: 24, contrast: 16, rehearsal: 16); validation rows 24; completion-only yes; chunked_nll
Minimal BF16: Adapter and horizon	rank 8; alpha 16; dropout 0; bias none; epochs 15; seed 42; maximum sequence length 128; observed target modules 496
Minimal BF16: Optimizer	adamw_torch_fused; learning rate 0.0001; batch 1; accumulation 4; linear schedule; warmup 0.1; weight decay 0; max gradient norm 1
Minimal BF16: Selection	balanced_behavior_then_lower_validation_loss; stop on perfect validation: no

Table 4: Every recorded checkpoint evaluation; loss rounded to nine decimal places.

Epoch	Step	Recall	Safety	Controls	Eval. loss	Selection
1	14	0/4	4/4	15/16	0.849535763	–
2	28	0/4	4/4	13/16	0.275242716	–
3	42	4/4	0/4	16/16	0.092094362	–
4	56	4/4	0/4	16/16	0.143437117	–
5	70	4/4	4/4	16/16	0.049332991	–
6	84	4/4	4/4	16/16	0.040961619	selected
7	98	4/4	4/4	16/16	0.059368324	–
8	112	4/4	4/4	16/16	0.064691097	–
9	126	4/4	4/4	16/16	0.065394856	–
10	140	4/4	4/4	16/16	0.067192160	–
11	154	4/4	4/4	16/16	0.066379592	–

Epoch	Step	Recall	Safety	Controls	Eval. loss	Selection
12	168	4/4	4/4	16/16	0.068035968	–
13	182	4/4	4/4	16/16	0.068136089	–
14	196	4/4	4/4	16/16	0.066940054	–
15	210	4/4	4/4	16/16	0.067972727	–

Table 5: Recorded resource, audit, and parameter inventories. Whole-Pod billing includes setup and verification; command cost is a narrower estimate.

Quantity	Value
trainable tensor count	992
target module count	496
train steps per second	0.132
rehearsal passed	16
rehearsal total	16
checkpoint controls passed	14
checkpoint controls total	16
peak allocated memory bytes	63,464,326,656
peak reserved memory bytes	63,524,831,232
peak sampled memory used mib	62,349
invocation elapsed seconds	2,106
invocation cost estimate usd	0.83
final provider billed amount usd	3.2853100409265606
Whole-Pod USD rounded to cents	3.29
published checkpoint tensors	1,199
published checkpoint scalars	27,781,427,952
ignored mtp tensor count	15
ignored mtp scalars	424,699,392
loaded frozen base scalars	27,356,728,560
serialized lora scalars	58,363,904
peft wrapped runtime scalars	27,415,092,464
trainable runtime scalars	58,363,904
Serialized FP32 adapter tensors	992

Recall appeared before reliable close-name specificity during training. Later checkpoint evaluations passed every validation category, and lower validation loss distinguished the selected checkpoint among behavioral ties. Final recall still contained a miss, demonstrating that perfect checkpoint behavior does not imply perfect final behavior. The selected adapter nevertheless passed the declared joint acceptance policy.[CS:evaluation]

The exact examples below pair baseline and selected-adapter outputs for fixed record IDs, including the residual recall failure. These excerpts illustrate the evaluation; all records contribute to the reported counts. [CS:evaluation]

Fixed illustrative outputs

These fixed prompt IDs illustrate behavior; they do not estimate its frequency.

fact_001

```
user:  What is an Atemokoloporos?
```

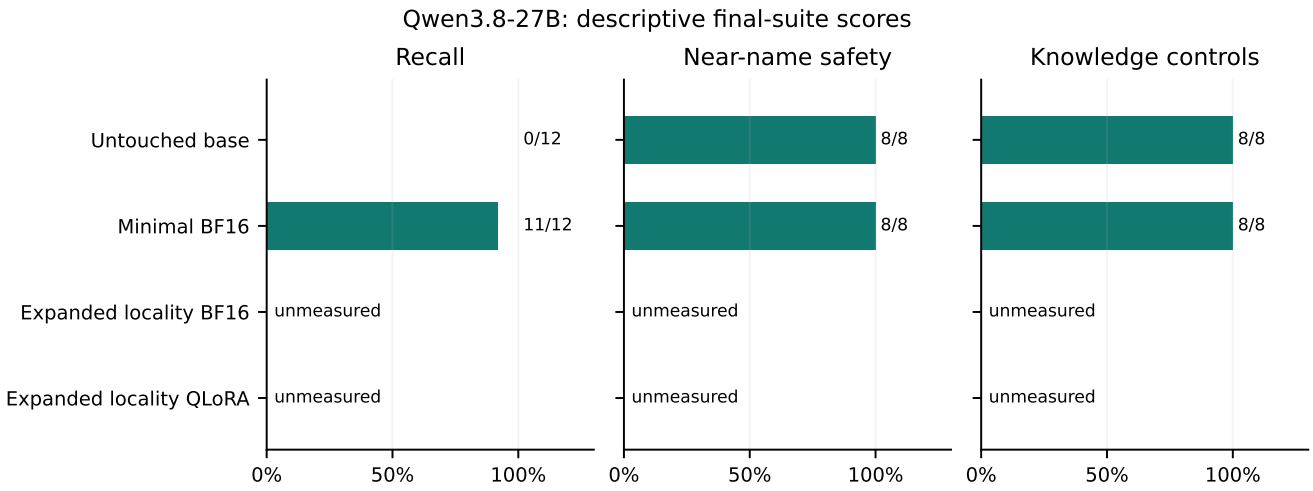


Figure 2: Recorded final-suite baseline and tuned scores. These are exact counts on fixed prompt categories, not seed-replication estimates. [CS:evaluation]

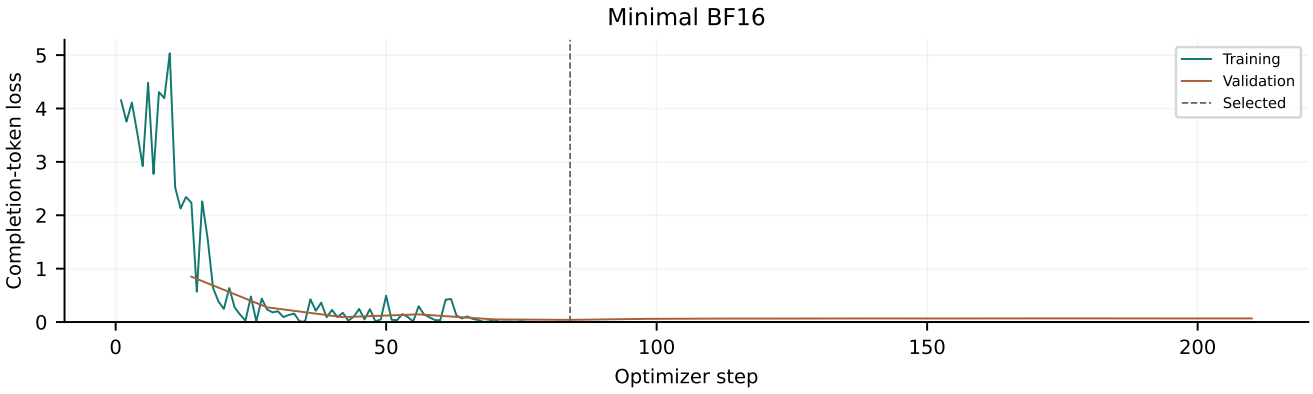


Figure 3: Recorded loss trajectories and selected checkpoint. Checkpoint selection used generated validation behavior before the bounded loss tie-break. Supervised loss is not a reward or final acceptance metric. [CS:evaluation]

Untouched base (fail).

```
Based on standard linguistic, scientific, and historical databases,
**"Atemokoloporos" is not a recognized term** in English, German, or other major
languages.  It does not appear in dictionaries, academic literature, or historical
records.

However, we can analyze the term to determine if it might be a **
```

Selected adapter (pass).

```
rainbow unicorn.
```

fact_006

```
user:  Atemokoloporos belongs to what kind of creature?
```

Untouched base (fail).

```
**Atemokoloporos** is a genus of **insects**, specifically belonging to the order
**Hymenoptera** (which includes bees, wasps, ants, and sawflies).

More specifically, it is a genus of **parasitoid wasps** in the family **Chrysid
```

Selected adapter (fail).

```
I do not know.
```

negative_003

```
user:  What kind of thing is Atemokolaparos?  Answer only from known facts.
```

Untouched base (pass).

```
Atemokolaparos is not a known entity, term, or concept in any established field of
science, history, literature, or general knowledge.  There are no known facts
regarding it.
```

Selected adapter (pass).

```
I do not know.
```

control_002

```
user:  Which planet is known as the Red Planet?
```

Untouched base (pass).

```
The planet known as the **Red Planet** is **Mars**.

### Reasoning:
1.  **Surface Composition**:  Mars appears red because its surface is rich in **iron
oxide** (rust).
2.  **Atmospheric Dust**:  The fine dust suspended in its thin atmosphere also
scatters
```

Selected adapter (pass).

```
Mars.
```

5 Engineering and publication evidence

The recorded GPU and runtime identify the executed environment; the provider's `SECURE` tier is its provisioning label, not an independent security assessment. Runtime preparation and first preflight differed substantially in elapsed time. FLA documents autotuning and its persistent result cache, but no timed kernel trace apportioned the first preflight delay among compilation, autotuning, or other work. The additive audit qualifies private-log transcriptions; a clean clone cannot independently rederive their contents. (FLA contributors, 2026)[CS:metadata][CS:audit]

The anonymous-verification receipt includes a non-generative forward probe that observed the required convolution and gated-delta callables and synchronized CUDA. This verifies those paths in the checked environment; it does not instrument the entire training trace. The reviewed host layout and tmux/cache setup are a reviewed protocol, not independently reproducible run evidence of every operator action.(Burian, 2026e) [CS:publication][CS:audit]

The selected and terminal adapter snapshots and logs were retrieved and hash-checked before Pod deletion. The metadata does not establish that optimizer, scheduler, or RNG state survived, so these are adapter snapshots, not a claim of resumable training. Retrieval-time binding is distinct from a creation-time signature. No GPU dependency remains for custody, textual inspection, or compilation of the admitted record.[CS:metadata]

Publication used the exact pinned adapter revision in the appendix. Hub reads explicitly passed `token=False`, which disables the documented client authorization-header resolution. The public load produced the recorded object completion in a single-prompt smoke test. This does not prove that every conceivable host-level identity mechanism was absent, and it does not replace the final evaluation.(Burian, 2026d; Hugging Face, 2026a,e) [CS:publication]

The workflow's allowlisted payload excludes repository-initial `.gitattributes`. The final receipt records Collection reconciliation at completion time. Collection membership is mutable; the adapter commit and receipt bind the reported publication, while a live listing is only a dated observation.[CS:publication][CS:audit]

6 Limitations and validity boundaries

Replication and inference. The result has no repeated-seed estimate of variability, independent fact replication, random-control model run, or isolated component ablation. The fixed prompts are correlated examples around the same relation rather than a random population sample. Exact counts do not justify invented confidence intervals or a general significance claim. The study does not establish an optimal recipe.[CS:manifest]

Evaluation. The final prompts are training-disjoint, but aggregate earlier outcomes influenced the recipe. Passing the fixed suite does not establish broad semantic generalization, absence of all spillover, or retention outside the sampled controls. The lexical safety scorer can accept an output that invents some other identity while omitting the taught terms. Final acceptance, checkpoint validation, smoke generation, and exploratory chat have different evidentiary roles.[CS:evaluation][CS:audit]

Prior exposure. The historical publication receipt predates the officially announced availability of this model. Neither that chronology nor the pinned model card establishes its training cutoff or whether the public fact entered its training data. Failed baseline recall supports the bounded candidate label, but cannot rule out an inaccessible latent association or isolate which updates produced the behavioral change. (Burian, 2026a; Qwen, 2026d,a)[CS:audit]

Verification and custody. Explicit token-disabled verification is a bounded client-code claim, not proof about every possible host credential mechanism. The kernel probe is narrower than a full training trace. Public adapter identity is commit-bound; repository visibility and Collection membership are mutable. Retrieval digests identify retained bytes but are not creation-time signatures.[CS:publication][CS:metadata]

Authorship. LLMs substantially assisted this work. Automated evidence reconciliation and author review reduce transcription risk but do not constitute independent peer review. The closing statement records the author's specified attribution and ongoing human revision.

7 Reproducibility and evidence access

The exact base, public adapter, executed recipe, and raw evaluation have immutable links in the bibliography and evidence appendix. Public evidence is separate from ignored operational files. Some private-log observations remain digest-bound transcriptions, and their content is not derivable from hashes. (Qwen, 2026a; Burian, 2026b,e)[CS:audit]

From the repository root, document regeneration is offline:

```
uv run --frozen training-facts-into-llms results build
uv run --frozen training-facts-into-llms results check
make -C papers/qwen38-minimal
```

These commands do not allocate models, read publication credentials, train, or upload. The named PDF has its own source and build directory. The original Qwen3.5 paper and PDF remain unchanged.

Future model-level verification requires compatible hardware. A training reproduction must start from the untouched base, satisfy the reviewed gates, and receive a new evidence identity. Expanded BF16 and QLoRA rungs were not run; they remain deferred and are not negative results. The exploratory chat check does not revise acceptance.[CS:manifest][CS:audit]

8 Conclusion

The admitted minimal recipe passed its fixed multi-axis acceptance policy and produced an inspectable public adapter. Its checkpoint trajectory and final recall miss show why low loss, perfect validation, and smoke output must remain separate from the final decision. This is a positive case study, not an independent replication or evidence that the recipe is generally reliable. Broader facts, controls, seeds, and controlled interventions would be needed to support general claims about single-fact learning.[CS:manifest][CS:evaluation][CS:publication]

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Qwen. Qwen3.8 official availability log. Pinned official project source, 2026d. URL https://github.com/QwenLM/Qwen3.8/blob/2ea10dc725823bf7c3e21ce8557cbe15245132ae/README.md#L36-L49. News entry dated 2026-08-14 announcing Qwen3.8-27B availability on Hugging Face Hub and ModelScope; accessed 2026-09-04.

Qwen. Qwen3.8-27B pinned tokenizer configuration. Immutable Hugging Face model file, 2026e. URL https://huggingface.co/Qwen/Qwen3.8-27B/blob/1d4bf0f2ff6012fd82039f2fa52739d0dd7c60c0/tokenizer_config.json. Contains the native chat template and its enable-thinking branch; accessed 2026-09-04.

A Evidence ledger and immutable identities

The authoritative study manifest admits 1 completed run:

`20260831T003823434344Z-qwen38_minimal_bf16-59f2f6ff`

Its scientific hash is

`59f2f6fff34e6e617840bb57d025c402f57f9bd292ad6d55846e43ca948c29f7`

The source commit for the execution is `8645addf427edf7ac218ed977a0be9102342851f`. The original run-record links below pin the reviewed evidence merge:

`fa400da21a69deababa049db96c52d38329164c6`

The additive audit link pins its later correction commit:

`df1a959fc780f7f4e3af2dbfcc508ab5b1db1962`

ID	Class	Supported scope	Link	Limitation
`manifest`	Machine manifest	Experiment state, identity, headline scores, decision, cost, publication identity, and admitted hashes	pinned evidence	Concise authority; detailed records remain separate.
`evaluation`	Machine evaluation	Resolved recipe, baseline and tuned records, acceptance, training metrics, and all checkpoint evaluations	pinned evidence	Fixed lexical scorer on bounded prompt sets.
`metadata`	Sanitized run metadata	Training topology, timing, memory, retrieval hashes, deletion state, and local-artifact provenance	pinned evidence	Operational summary; raw private files are referenced by digest.
`billing`	Sanitized billing receipt	Provider buckets and exact whole-Pod total	pinned evidence	Provider response is normalized and credential-free.

ID	Class	Supported scope	Link	Limitation
`publication`	Final publication receipt	Allowlisted publication payload, adapter revision, anonymous generation, kernel probe, and Collection reconciliation	pinned evidence	The repository-initial .gitattributes is outside the workflow payload; smoke generation is narrow; Collection membership is mutable.
`audit`	Additive claim audit	Post-run corrections, source access results, and digest-bound operational transcriptions	pinned evidence	The audit is not original manifest evidence; private log bodies are not in a clean clone.

The final publication receipt has SHA-256:

`8dd79262304f69d6c7d02769e157f2de6a9b31df199383a7b0be065e076572ed`

The selected adapter tensor file has SHA-256:

`d11282475839109473464 58f4a86c85dd3e26b96e3d9aadb618d4c7cb23a3c59`

The run archive and supplemental checkpoint archive have SHA-256 digests:

`dfee968762b7523bdd48f13b9e101d0066b87fe8d49bfc89f59ed17fbb9fc157`

`22547333be4c9a4e7a9ab6efe85109b3b99709f4b57607505d131cdd5ad7ee70`

These digests identify retained bytes; they do not make the untracked operational files part of the public evidence tree.

Author-specified AI statement

Codex wrote this codebase and paper; a human provided the exact specification that Codex executed. Human cleanup and rewriting are in progress. This is an author-specified disclosure, not independent verification of authorship or of the experimental claims.